

Sustainable Waste Management



Sustainable Waste Management

Introduction to Waste Management:

- Waste is generated from households, industries, agriculture, healthcare, and construction activities.
- Rapid urbanization and population growth are increasing waste generation worldwide.
- Poor waste management causes pollution, disease, and environmental degradation.
- Sustainable waste management aims to minimize these negative impacts while conserving resources.



Sustainable Waste Management

What is Sustainable Waste Management?

- Sustainable waste management means managing waste while minimizing its environmental and societal impacts.
- It considers the entire process, from waste generation to final disposal.
- The approach aims to use materials and resources more efficiently.
- It emphasizes waste prevention, reduction, reuse, recycling and recovery before disposal.
- It supports the broader goals of environmental protection and sustainable development.



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Key Features of a Sustainable System:

- 1. Resource Efficiency**
 - Use materials only when necessary.
 - Avoid unnecessary consumption of resources.
 - Improve efficiency during production and construction.
- 2. Waste Reduction**
 - Minimize waste generation at its source.
 - Reduce unnecessary packaging and material use.
- 3. Sustainability**
 - Consider environmental, economic and social consequences.
 - Plan waste management over the long term.

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Sustainable Waste & Resource Management Practices:

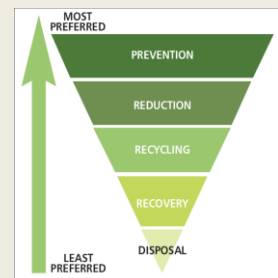
Important practices include:

- Proper planning before material use and waste generation.
- Optimum utilization of materials to reduce unnecessary waste.
- Waste minimization during production, construction and consumption.
- Proper management of construction and demolition waste.
- Greater use of reclaimed and recycled materials.
- Providing adequate recycling space and facilities.
- Encouraging cooperation among producers, consumers and authorities.

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Waste Management Hierarchy:

The hierarchy provides a preferred order for managing waste:



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Waste Prevention and Reduction:

Waste Prevention:

- Prevent waste before it is created.
- Consider waste reduction during **product design and manufacturing**.
- Design products for longer use and easier reuse.

Waste Reduction:

- Use fewer raw materials.
- Reduce unnecessary packaging.
- Improve production efficiency.
- Extend product life.



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What is reuse?

Reuse means preparing an existing waste item or component so that it can be **used again**.

This may involve:

- Checking the condition of the item
- Cleaning
- Repairing
- Refurbishing
- Reusing whole products
- Reusing spare parts



Why is reuse important?

It keeps materials in use and reduces the amount of waste requiring recycling, recovery or disposal.

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What is recycling?

Recycling involves:

Collection → Separation → Processing → New product/material

Waste materials are processed to produce new substances or products.

Examples include recycling **paper, glass, metals and plastics**.

Newspapers can be recycled into new newspapers or insulation materials.

Recycling reduces the demand for new raw materials.

Successful recycling requires participation from **both producers and consumers**.

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Composting:

Composting as Organic Waste Recycling

- Composting applies recycling principles to **organic waste**.
- Common materials include food waste and other biodegradable materials.
- Organic materials are biologically transformed into **compost**.
- Compost can be used to produce or improve fertilizer products.
- Composting reduces the amount of organic waste sent for disposal.



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Energy Recovery:

Energy recovery is used when recycling is **not possible or appropriate**.

Major processes

- Anaerobic digestion
- Incineration
- Gasification
- Pyrolysis

Possible outputs

- Fuels
- Heat
- Electricity
- Other useful materials

Energy recovery can reduce the volume of waste requiring final disposal while recovering useful energy from waste.



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Disposal: The Least Preferred Option

Disposal is the **bottom level of the waste hierarchy**.

Main disposal approaches

- **Landfill:** waste is placed in a designated disposal facility.
- **Incineration without energy recovery:** waste is burned without recovering useful energy.

Why is disposal least preferred?

- Useful materials are lost.
- Additional land may be required.
- Disposal can create environmental impacts.
- It does not preserve the resource value contained in waste.

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Environmental Benefits of Sustainable waste management :

Sustainable waste management can contribute to:

- **Cleaner and healthier environments**
- Conservation of **natural resources**
- Conservation of **energy**
- Reduction of greenhouse-gas emissions
- Reduced dependence on landfill
- Reduced demand for extraction of new raw materials
- Lower environmental impacts associated with material production and disposal

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Economic and Social benefits of Sustainable waste management :

Economic benefits

- Lower waste collection and disposal costs.
- Reduced energy consumption in manufacturing.
- More efficient use of raw materials.
- Development of markets for recyclable materials.

Social benefits

- Cleaner surroundings.
- Greater public awareness of responsible waste management.
- Increased cooperation among communities, authorities and waste-management organizations.
- Potential employment opportunities in recycling and resource recovery.

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Challenges to Sustainable Waste Management

Major challenges include:

- Limited availability of **trained professionals**.
- High transportation costs.
- Inefficient waste collection and transportation systems.
- Inadequate budgets.
- Poor equipment management and maintenance.
- Uncontrolled scavenging.
- Low public awareness.
- Lack of alternative treatment strategies.
- Illegal burning and dumping of waste.

Thank you all
